Frontend Architecture

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Frontend Architecture

- Prepare
- Scope
- Model
- Develop
- Design System

UI Components & State

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Develop

Frontend Architecture

- Design System
- UI Components & State
- Data
- Render & Optimize
- Secure

Test

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Render & Optimize

Frontend Architecture

- Secure
- Test
- Run
- Build
- Monitor

Deploy

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Secure

Frontend Architecture

- Test
- Run
- Build
- Monitor
- Deploy

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Prepare

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Prepare

- IDE
- Terminal Emulator
- Shell

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IDE

- Cursor
- Visual Studio Code

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Terminal Emulator

- Ghostty
- iTerm 2

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Shell

- Z Shell
- Bash
- DevEnv

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Scope

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Scope

- Requirements
- Constraints

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Requirements

- Non-Functional
- Functional
- User Journey
- Accessibility (a11y)
- Internationalization (i18n)

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Requirements

Functional

- User Journey
- Accessibility (a11y)
- Internationalization (i18n)
- Localization (L10n)

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Constraints

- Compliance
- Stakeholder Needs

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Model

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Model

- User Flow
- Data Models

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Data Models

- Entities
- Schema Validator
- API Interfaces

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Entities

Attributes

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Schema Validator

Zod

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Develop

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Develop

- Component Patterns
- Design Patterns
- Linting & Formatting
- Architecture Styles
- Repo Management
- Package Management

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Linting & Formatting

Develop

- Architecture Styles
- Repo Management
- Package Management
- Documentation Tools
- Collaboration

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Component Patterns

- Container/Presenter
- Higher-Order Components
- Hooks

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Design Patterns

- Proxy
- Adapter

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Linting & Formatting

- ESlint
- Prettier

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Architecture Styles

- Islands Architecture
- Micro-frontend
- Monolith

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Repo Management

Turborepo

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Package Management

- npm
- pnpm

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Documentation Tools

- Storybook
- JSDoc

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Collaboration

- GitHub
- JIRA
- GitLab

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GitLab

Revisit GitLab offerings/capabilties

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Design System

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Design System

- Components
- Styles
- Development
- Strategy

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Components

- Unstyled primitives
- Frameworks
- Styled primitives
- Copy/Paste

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Unstyled primitives

- React ARIA
- Radix UI

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Frameworks

- DaisyUI
- Chakra UI

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Styled primitives

Shadcn

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Copy/Paste

Aceternity UI

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Styles

- CSS-in-JS
- CSS

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CSS-in-JS

- Restyle
- StyleX
- CSS Components

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CSS

- Tailwind
- Open Props
- CSS Modules
- Frameworks

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Development

Storybook

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Strategy

- Atomic Design
- Token-based Design
- Component-driven Design (CDD)

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Token-based Design

- Design Tokens
- Themes

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Design Tokens

Style Dictionary

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Themes

Light/Dark Mode

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UI Components & State

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UI Components & State

- Frameworks & Libraries
- State Management
- Data Flow

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Frameworks & Libraries

- React
- Svelte
- Vue
- Web Components
- React Native

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Frameworks & Libraries

- Svelte
- Vue
- Web Components
- React Native
- Angular

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React

Next.js

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Svelte

SvelteKit

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Vue

Nuxt

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Web Components

Lit

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React Native

Expo

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State Management

- Global
- Local
- Browser
- Async

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Global

- Redux
- MobX
- Zustand

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Local

- Component State
- Context/Provider
- Signals
- Micro State

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Micro State

- Jotai
- Zustand

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Browser

LocalStorage

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Async

- Fetch state
- Caching
- Background/Offline

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Fetch state

- isLoading/isError/isSuccess
- Optimistic Updates

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Caching

- Client-side
- In-memory
- Browser

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Browser

LocalStorage

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Background/Offline

Service Workers

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Data Flow

- One-way binding
- Unidirectional
- Observables

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Data

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Data

- Fetching
- Modeling
- Transmission
- Form
- Transformation & Mapping

Caching

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Fetching

Data

- Modeling
- Transmission
- Form
- Transformation & Mapping
- Caching

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Fetching

- Undici
- window.fetch

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Modeling

- MobX State Tree (MST)
- Zustand Tree

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Transmission

- REST
- Server Sent Events (SSE)
- Realtime
- GraphQL

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REST

- Express
- Koa
- Hono

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Realtime

- WebSockets
- WebRTC

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Form

- JSON Forms
- react-jsonschema-form

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Caching

Tanstack Query

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Render & Optimize

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Render & Optimize

- Build Performance
- Client-side Performance
- Template Engines
- Strategies

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Client-side Performance

- Critical CSS
- Lazy-loading
- Code splitting
- Hydration
- Pre-fetching

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Client-side Performance

- Hydration
- Pre-fetching
- Caching
- Image Optimization
- Core Web Vitals

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Template Engines

- EJS
- MDX
- Handlebars

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Strategies

- Client-side Rendered (CSR)
- Server-side Rendered (SSR)
- Server-side Generated (SSG)
- Progressive Web Application (PWA)
- Streaming SSR/SSG

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Strategies

- Client-side Rendered (CSR)
- Server-side Rendered (SSR)
- Server-side Generated (SSG)
- Progressive Web Application (PWA)
- Streaming SSR/SSG

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Secure

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Secure

- HTTP
- Browser Security
- API Security
- Data Security

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HTTP

- HTTP Strict Transport Security (HSTS)
- Cross-origin Resource Charing (CORS)

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Browser Security

- Cross-site Scripting (XSS)
- Cross-site Request Forgery (CSRF)
- Input Validation
- Content Security Policy (CSP)

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API Security

- Authentication Mechanisms
- Authentication Providers
- Authorization Mechanisms

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Authentication Mechanisms

- Password
- OAuth
- Passwordless
- Multi-factor & 2-Factor (MFA/2F)
- Single Sign-on (SSO)

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Authentication Mechanisms

- OAuth
- Passwordless
- Multi-factor & 2-Factor (MFA/2F)
- Single Sign-on (SSO)
- Token-based

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Authentication Providers

- AWS Cognito
- Firebase Auth
- Clerk
- Auth0

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Authorization Mechanisms

- RBAC (Role-based)
- ABAC (Attribute-based)
- Roles & Capabilities

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Data Security

- Encryption
- Compliance

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Test

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Test

- E2E
- Integration
- Unit
- Mocks
- Coverage

Testing Framework

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E2E

Test

- Integration
- Unit
- Mocks
- Coverage
- Testing Framework

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E2E

- Playwright
- Cypress

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Unit

- Vitest
- Jest

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Testing Framework

- Testing Library
- Mocha

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Run

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Run

Runtime/Backend

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Runtime/Backend

- Deno
- Node

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Build

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Build

- Bundle & Package
- CI Pipeline
- Compile
- Cloud Environment

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Bundle & Package

- Vite
- Rollup
- Webpack

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CI Pipeline

GitHub Actions

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Compile

- Turbo
- Esbuild

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Monitor

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Monitor

- Observability
- Logging
- Metrics

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Observability

- Cloudwatch
- Datadog
- New Relic

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Logging

Pino

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Metrics

- Sentry
- Grafana
- Real User Monitoring (RUM)

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Deploy

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Deploy

- CD Pipeline
- Platforms
- Delivery
- Strategies

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CD Pipeline

- Automation
- CI/CD Tools

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Automation

Build/Test/Deploy

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CI/CD Tools

GitHub Actions

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Platforms

- Vercel
- Deno Deploy
- AWS
- Hosting
- Edge

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Platforms

- Vercel
- Deno Deploy
- AWS
- Hosting
- Edge

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Hosting

- Dynamic
- Static

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Dynamic

- Kubernetes
- Docker
- AWS Lambda
- Serverless

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Static

- Vercel
- Netlify
- AWS S3
- AWS CloudFront

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Edge

- Cloudflare Workers
- Vercel Edge Functions

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Delivery

CDN

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Strategies

- Roll-back
- Canary
- Blue/Green
- A/B Test

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Thank you

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